

Mahault Albarracin ~ Active Inference Insights 002 ~ Epistemic Communities, Social Scripts, AI

Source: [YouTube](#) **Playlist:** Active Inference Insights **Duration:** 1:47:18 |

Uploaded: Unknown **Transcript method:** auto_caption

hello everybody and welcome back to the active inference insights podcast brought to you by the active inference Institute I'm your host Daris parvy Wayne and today I have the pleasure of chatting to ma Al barassin ma is a cognitive Computing PhD candidate at the University of kbec in Montreal as well as the Director of product Innovations and research at verses a company which leads the way in applying the active inference framework to AI with expertise in computational Neuroscience sexology Consciousness and much more M's work is Broad Innovative and deeply exciting which makes me rather um excited for this conversation ma welcome to the show thank you so much for joining us it's a as I said it's an absolute pleasure well thanks for having me so um yeah as I sort of mentioned in the introduction it your work covers a very broad range of topics um and areas of research but as you know this podcast is exploring the themes of active inference and the free energy principle so I want to mainly focus on that even though we can see about how these different themes tie in um with that overarching um framework if one goes on your Google Scholar and sees your work on active inference um again they'll see it's diverse where I want to start is the paper that you wrote on epistemic communities um I guess the first question our audience might have um just as a reminder for those listening this podcast is very much meant as a gentle introduction to active inference um as gentle as an introduction to active inference can be um so just for our audience my first question is very simple and it's just what is an epistemic community so it has many definitions uh it's it's a Loosely used term in variety of fields but the way we've used it is essentially a community of people that share very specific types of beliefs that lead them to sample reality in the same way so if you think about like um your Norms they usually are tied to a specific way of thinking like for instance the fact of going to work the fact of going to work 9 to 5 is tied to an ideology that values a certain type of productivity consumerism Etc and a certain type of relationships where we are together in the morning together in the evening and um we are with different people during the day etc etc etc so your life becomes

shaped by these ideas that give rise to Norms cool and um let's sort of tie in the active inference framework then to this idea that we have a community which is tied together through um sampling similar Norm similar sensory observations um perhaps you could start just by giving us a brief uh or yeah relatively accessible um notion of what active inference is I mean KL did a very good job last week at explicating this maybe just as a reminder to people um and then how that framework um and the truths that the free energy principle discloses can be applied to the formation of an community yeah so active inference um as a principle is rather simple it's really just the idea that we try to minimize how much we are likely to dissolve we don't want to die so in order to do so we're going to try to map the world and understand what might be coming such that we can avoid moments where we might be dissolving or moments that are dangerous for us more surprising we try to uh create evidence of ourself we try to predict ourself existing in the future um and to do this we um make a prediction act in the world try to change something see if it changes the way we thought it was going to change and then update our model of the world so you got this uh sensory active Loop where uh the agent is embedded in the world rather than just outside of it mapping it so it's a pretty crucial idea that the agent necessarily has a perspective which is defined by how they are separate and yet embedded in the world they have a boundary we generally refer to that as a mark blanket this boundary is both defining and defined by the outside world and the agent themselves um so if you understand that the agent necessarily has to map the world in order to predict what's likely to happen but that the agent is only capable of dealing with their perspective and how much they can compute from their perspective you understand the agent is fundamentally limited it has to um to maximize how much information it has about the world to communicate with potentially other agents which will also be gathering information about the world not just from each other but from their own experience within the world now now what we posited when we were doing our research is that it would be more costly to just flat out change your entire model of the world if you are wrong then try to seek out agents who might validate your view of the world and therefore also share the same model this is where we bring in the notion of Niche construction when agents exist in the world as I said they evidence their own existence to do so they shape the world so anytime I walk on the grass I make a little dent and I create a little path and so if other agents like me same agents with the same model are going to take the same path eventually that path is going to be pretty obvious it's going to pull other agents to that path so I and other agents like me are shaping the environment to resemble Le us so we therefore thought that confirmation bias here was a way

for agents to avoid having to constantly change their model and were more likely to be with people like them create a world that resembles them and eventually create something that resembles an echo chamber um at least on the belief on the belief path what we also posit is the outcome of that is you're not just sharing beliefs you're creating the world around you such that eventually some communities will in fact create some degree of the beliefs they are sharing in the world around them which is no longer at all the same world as some other community which may be sharing together another model and they it becomes practically incommunicable not only are they not sharing the same model they're not sharing the same world they're not sampling it the same way and so communicating here is essentially impossible amazing that's that's super interesting um and I think definitely maps on to people's experiences in um if they've ever found themselves down the rabbit hole of an echo chamber whether that's on Twitter or Instagram or wherever I guess a question that comes to my mind here is that um maybe a um heuristic idea would be that truth exists as some kind of Prior within um a generative model that an agent like us has like it's important that we tie ourselves in some way to um veracity because if not then we're going to be constantly surprised every time something false happens or or that false true distinction it's a bit blurry but but you see where I'm going so a lot of EO yeah yeah so a lot I'm going with this is that a lot of echo Chambers are reinforcing their model of the world but it might not necess it doesn't necessarily need to subscribe to any notion of external truth to what degree does that ever butt up against the a kind of inherent prior which is that eventually the falsity of your claims is going to come home um or can or can the echo chamber the entrenchment just keep going keep digging as far down as as possible yeah so this is a super interesting idea and we first have to Define what we mean by truth and what level of Truth what do we mean by truth do we mean something that I can observe do we mean something that is verifiable by other agents uh and then what level of Truth do we mean um the truth of the event the social truth of the meaning of the event um the the fact that to some extent no matter what happens this element will be considered true do you see what I mean like there's so many ways in which this can break down and what this leads us to is so this is a theory this is something I'm trying to develop obviously it's not uh verified yet so it's still something to to take with a few grain of salt but it's it's a theory that I think U naturally follows from what we've done depending on the level of truth you're talking about and your definition of Truth the fact that something say happened or not can be irrelevant what matters about truth is whether it's actionable to some extent so let's take an example if I decide I can fly but I never act on it it has no impact on my

life my sensory active Loop isn't going to change my model is not going to change so I can carry this belief without ever having any consequence at that level I'm not going to die but now I can tell other people that I Can Fly And if none of us act on it but we all squarely believe we can fly and it serves a social purpose of identifying each other we know who we are and then we congregate to each other we create a reality together we create purposes what happens then where does the buck fall where where does that belief become a belief that potentially harms us I think that's the true question so think of things that are more contentious like does God exist technically there is no way to prove it literally part of the definition is that believing belief in God has to be faith-based or at least you know for Christian judeo-christian beliefs it has to be faith-based which means by definition you shouldn't need evidence you should believe without evidence you should take a leap of faith okay then we by virtue of that we can't verify it we can't change it your beliefs is kind your belief is kind of impervious so then what are the outcomes comes of that belief how does it allow a community to survive how does it allow people to survive and there's a really interesting phenomenon that happens there some people will start manufacturing evidence of God they will start think okay well yes it's true I don't need evidence but I can read certain kinds of things as evidence and they will select the kinds of things that count as evidence I.E things that they can't ascribe a clear cause to so for instance they won't ask God to feed their dog because if they did and the dog doesn't get fed the dog dies and now they're like well God didn't do it there's a Clear Proof here it didn't happen but something more nebulous like in the future I will get a job I mean you know like it's highly likely you're GNA but can you really describe the cause of that like you might have sent some resumé's but there's no guarantee anyone will pick up your resume like it's it's practically luck at this point so if it's luck it's God because luck is not something we can ascribe a clear cause to um so so then you get these communities that are functional because that belief never really comes into play in their actions and so you can have weird types of layered beliefs um for certain people like you could have a flat earther be your manager it probably won't change much it's not compatible with your beliefs but in terms of the the layer that you share the functional layer that you share it doesn't come into play but if I Believe I Can Fly And therefore you can fly and I tell you well go go do it now fly jump off the building now we're going to have a problem and therefore that kind of belief is less lik to be carried because functionally it's likely to disappear self in terms of self-organization it just is likely to disappear right yeah that's interesting I guess it's it's absolutely right to unpick these different layers and the fact that I guess one way of looking at it is that truth can have this kind of

instrumentalist or pragmatic aspect to it but also this um something that you know philosophers of language would think about a little bit more in terms of conditional statements and whether something's actually true in the world I wonder whether therefore if we look at the Active inference framework and we try to apply Act of inference to this a question that comes up in my head is does for an agent to consider their belief to genuinely be true because I think what we can say here is if I had the belief that I could fly perhaps I occupy that truth at an instrumental level because I'm surrounded by other people who believe that they can fly and it actually gives me a sense of purpose and and and and so I almost delude myself into thinking that but really if push came to shove and someone had me at the top of a cliff Edge I probably wouldn't do it because I don't think there's actually a state of the world out there which says Darius can fly does the all of that to say does ACT does the active inference framework require for an agent to believe that there is genuinely solid epistemic grounds to believe that something is ontologically the case so that I can actually fly that if I took off that I would need to be able to sample or have sampled evidence for that or is there a mechanism that might be um peripheral to that which could give me the um trust in a sufficient trust in my model that I maybe I would jump off the cliff so I think we have to look at how um life is realized and how you plan yourself in the future right so let's assume you really think you can fly and your life just doesn't lend itself to it like this how you do you take the car you you know you hang around you you have coffee you work whatever it's just it just never comes into play you could hold that belief with high certainty because it never gets tested and there's never a risk now if you were to start having a real risk like that it it actually like that's that state is not highly unlikely it's like it's over here and you now have to start testing your own predictions of yourself into the world and you're going to start seeing spikes of free energy like there's like you've never seen anyone fly you sure you can but you've never seen anyone fly and the risk associated with not flying is really really really high um I think that's when you start seeing the armor break like I think the difference becomes between someone who is lost touch with actual reality like they genuinely believe they can fly and it becomes an actual thing they need to do now they probably will do it but they will also probably die which means that um these kinds of beliefs which we generally consider to be psychotic right like it's it's that's the definition it's a loss of touch with reality no matter what that means and so when we start couching these ideas in this way we start depathologizing a little bit or like well it's not they're not they're not sick per se they're nothing intrinsically wrong with them but their beliefs are no longer aligned with the ecosystem within which we exist if they were on say the

moon for a minute maybe you know they could fly like it wouldn't be an impossible belief it's just not a belief here that that works um so I think that's why most of the people who hold beliefs that are not uh related to what we would consider something that happened like it's not verifiable it's not uh it's not something that we can assure ensure happened or can happen generally these beliefs don't relate to things that are possibly um harmful to them like for instance believing in Trump believe in what what Trump says it's easy to debunk what he says like a lot of people have done it in fact it's actually it takes more time to debunk everything he says than just to listen to him but it has no impact on your life if you do believe him like pragmatically won't hurt you so it's it's it's um it's a very useful tool to lie about things that are completely unverifiable to you therefore why not believe them yeah I think um I think what you spoke about in terms of deep apologizing is very interesting when um I was speaking to Carl for those who are listening they might recall we we sort of spoke about the fact that the complete CL class faum states that actually any Behavior can be cast as um Bas optimal under some prior beliefs so I think that's a nice it's it leads itself it lends itself to a Humane approach to Psychopathology or confirmation bias less um you know more mundane instances of circular reasoning or or convincing yourself that something is true when it's clearly not um going back to your paper I thought an interesting thing was so so for those who don't know ma um and Maxwell and others sort of created a a computational model um in addition to the to the theorizing right so there is the theory and then you would put it in silico and you um you in dowed your silico agents with an a priori hypothesis that or or preference that they would be more likely to sample um agents who are similar to them which is in in a sense the kind of uh prior preference which underwrites um confirmation bias my question here is in that are you stipulating that we too have those prior preferences because on one hand yes I mean that manifests in the behavior and it makes sense in terms of confirming your own model at the same time humans can also be incredibly flexible creatures who do change their model on the basis of um evidence so firstly is the claim there that that is an actual um feature of human cognition and our generative model and two um if so would you couch that in terms of um natural selection and evolutionary pressures or is that is it even more uh fundamental to self-organization to um sample information from those who are more similar to you right so so confirmation bias was a parameter it was something we tested within with or on a Continuum and we saw what the effects of it were relative to the formation of echo chamber or to the change in beliefs um and then we saw that it had a magnifying effect but it wasn't as simple as just you turn it off and it's gone and you turn it on and it's it's there right there was still the

possibility to form communities without uh the notion of confirmation bias it just wasn't a strong and it also depended on the network Dynamics uh different networks Pro propagate information differently so um we we also had the notion of uh Habit formation so if you have uh the need to uh to have habits basically you're going to slowly start sampling the the same thing over and over and over again so um there there were a lot of parameters that we that we used and we found that uh um confirmation bias basically uh biased the an the agent's likelihood model to acquire more evidence about their environment if the information confirms their pre-existing beliefs it's it was that simple basically uh so basically it it just made it so that they would still sample people people around them but they were more likely to resample somebody that said something that agreed with their model which is extremely compatible with Act of inference right you're trying to self-evidence so you're going to try to sample things that seem to you know agree with your model um so confirmation bias was just a formalization of that specific parameter um what we see is it's actually adaptive for you to be with people like you when you are with people like you not only are you likely to understand people better and therefore share information more efficiently um you're also likely to be in an environment that is beneficial for you not just because you craft the environment around you but because if you find yourself with people like you you're probably finding yourself in an environment that benefits those people so you know it's it's just adaptive now it's a adaptive to an extent if you're only with people with exactly your your same model then you're also very likely to to um have the same issues and to be um vulnerable to the same kind of volatility it's like the it's like monoculture if you have one type of banana if there's one disease banana they're all going to get it and you no longer have bananas it's it's the same idea you want want to continuously grow and change and and and gather more error and increase your Fitness landscape such that you never get to a point where there's an error that none of you saw coming and it just wipes everything out like a Black Swan event um so there's like a a nice middle ground it is it's like a super stimulus if you find yourself in an echo chamber it feels really really good for time but there's a part of your model the part that wants exploration that eventually understands that you're stagnating and something needs to change something needs to happen so it's not because we try to confirm or model that we also just want the ma the model to be stationary we want to minimize free energy and we want to know that we are minimizing free energy right it's layered in metacognitive so in that sense you want to see yourself going down so if you're in a plateau you're no longer going down right um so that's why I think to some extent even when you have a group that seems to agree that seems to have

the same idea it'll eventually start disagreeing inside like there will be people that'll push that will keep polarizing such that now there's um a little bit of a difference like you're you're still some people like well no okay that's that's too far I I I don't think all of the World is Flat maybe you know a little bit but not all of it or I don't think the entire universe is just the planet that's silly do you see what I mean like it's the same belief it's the same kind um moving away from the trajectory that the others are going but you can't always constantly absolutely agree or you cease uh to learn information so you'll start again sampling a little bit outside it's just that if you're stuck in an echo chamber and the echo chamber has you know drifted drifted drifted drifted drifted so far away from anybody you could ever talk to such that there is no reconciliation Point here you're you're just going to have like a straight up no there's nothing to hold on to for you to be like well maybe my model is a little similar but not quite the same so I could you know adapt a little bit you're there's this this this Ridge is too far it's too it's too far gone so it is adaptive for you to be with people like you but if it's too like you then you're stagnating and eventually you're likely to potentially disappear amazing yeah loads of things uh come to mind from that answer um as you said again um for those listening as you said just important to point out as you said to reduce free energy over time so-called expected free energy in the actual mathematical formula there is this balance of epistemic and pragmatic actions so we can't we we do have to explore the world and we do have to sample in conditions of uncertainty so if people are curious this this sort of evades the so-called dark room problem which was a kind of an initial contention to the free energy principle which people can read about and it's over 10 years old now these arguments um I guess listening to this um something that comes to my mind is and well something that comes to my mind and maybe those listening is why does one end up in these states anyway so maybe I'm listening to this and I might I'm not but I might have been in some hypothetical World a flat earther and I might start panicking saying what does that actually say about me fundamentally that I was amenable to those claims whatsoever is there in some computational formulation or just some conceptual formulation an idea about why some people tend more than others to be wrapped up in conspiracy theory or group think um this monolithic way of um cognizing rather than cognitive flexibility and again I don't mean to put these two on sort of um binary polarities as if one is amazing and one's terrible as you say we do spend time with people we agree with all of us as as we should or else we'd have no friends um but is there a reason perhaps why some people end up really going down that rabbit hole and S you know and and and it's it's bizarre for those on the outside as to how they've got to that point right so let's always remember that

anything we're going to say is u a conjecture because social systems are extremely complex and it's a little bit like chaos theory like ascribing one or two causes to a very complex phenomenon is all always wrong we can have a model and it's an approximation but we can't make too many blanket statements so that's how I'm going to caveat what I'm about to say next the second thing I'm about to say is another caveat is everyone does group thing everyone does it like it's just that some group think is considered and that's what I meant by the truth yeah in end can you reproduce any of the experiments that come out from the scientific field can you like reproduce a proof can you run it yourself do you understand everything that comes out the truth is no you have to rely on others for an epistemic chain like is this a justifiable belief yes maybe I think you know that person looks really confident and other people seem confident in them so I'm you know I'm to follow that and sometimes what their model allows you to do his predictions about the world and those predictions happen to come true so you through abductive inference believe it must be true but it doesn't mean that in fact it could be completely unrelated maybe this is just like you know something that happens anyway if everybody told you you've got to jump and the sun will come up so we've jumped and the sun kept coming up and you're like all right I'll jump and you're like you're right the sun did come up wow that works so like it could be completely wrong but you still you know that's your best explanation so far it allowed you a prediction your prediction worked okay well there you go there's some evidence for that in in our case it's more along the lines of our beliefs are the right beliefs and our ideology is the right ideology because it leads to XYZ kind of kpi we've arbitrarily defined as the kpis that correspond to what allows us to minimize free energy but it's it's highly contentious often times the the kpis we Define are the kpis that allow us to stay in the state that we are and so it's heavily slanted towards people in power right propaganda media Etc all of these tools are built to get you to believe that you are correct and you are doing the right thing and that right thing is to give more money to rich people so you know let's let's keep that in perspective for the time being um why do some people fall down rabbit holes it's a very known thing that there are key um funnels for certain kinds of ideologies because certain kinds of ideologies have a dynamic they very quickly allow you to feel better about yourself feel better about yourself or or feel more in control of the world like you now have an answer you you can now understand what's going on your model feels like it reduce free energy pretty quickly um that's one of the reasons conservatism is often very popular or at least populist conservatism it tends to rely on very easy very simple answers to very complex problems why is your life crap immigrants obviously it's immigrants it's not the vast complex

industrial um uh series of Institutions that keep you in a position where everything around you has been eroded to the point where you have nothing to grab on to because that would be really complicated it's immigrants so we when we understand that certain kinds of patterns of Stark reduction of free energy exist exist in specific rhetorical uh patterns we then understand okay well who needs these kinds of reductions who are the people that are currently Feeling High uncertainty very vulnerable and need something to hold on to it's probably people whose model had evidence and sort of no longer does there is nothing in the world around them that continues to value validate their perspective so now they're searching give me anything anything that will allow me to continue this model of myself which has crafted who I am my phenotype depends on this model if it no longer exists now I have to accept that I am at the bottom of the food chain and I don't see a way out because it's complicated so who are the people likely to be like that well currently it's men who feel like they're in a crisis of uh of masculinity it's very poor people who feel like they're disenfranchised and nobody's helping them and um the only reason that would be the case is because the elites are um sucking the blood of babies and uh to do to to fulfill their evil ends to bring more immigrants in because the immigrants are the reason that you don't have jobs anymore etc etc etc like these conspiracy the theories generally tend to give you erroneous but easy answers and the problem is they're usually very wrong so they have to prop themselves up with these webs of Tangled beliefs which locally are a very easy answer right they're like why uh why do I see this well because you know that thing and that thing is conceptually related therefore that must be the answer and so we get into this really interesting space where there's people with incredibly complex webs of where you be like well that doesn't make any sense if it's more complex why would they hold on to this well because remember these things are layered they also get a sense of control from this they're minimizing free energy somewhere at the level of the easy little answer the easy little thing they got a small chunk of explanation it validates their position in the world and they now feel like they have control over things they understand the world it's actionable so all of these reasons give you a very clear profile for the kind of person who is likely to fall for that pattern of beliefs fascinating okay good that's a yeah yeah that's that's really useful um and yeah perhaps good to remind people that um in terms of the actual formula for the free energy Principle as you know it's can be cast as accuracy minus complexity of your model so always seeking the so-called most accurate accurate model in the sense it fulfills our preferences without added layers of complexity about greater degrees of freedom and maybe to to flesh out what you were saying you're right people have these

incredibly complex models of how the world works and it's a pedophile ring and that's somehow linked to 911 and them but they kind of internally cohere in some strange Nar narrative way they're not complex in the sense that they involve contradictions because if they did that would be the buildup of prediction error so it's like you have this stacked uh generative model but because everything kind of contextualizes and constraints everything at the lower level there's actually minimal prediction ER to reduce it's just that if you actually spit that out to an external World it just it turns out to be [] **and I think that kind of explains why some of these people have massive existential crisis at some point if they're just like oh []** that whole thing was really really wrong because the whole edifice comes crumbling down and you get this enormous Spike of prediction error every layer um and there's something interesting too like even when they encounter a contradiction they have the power to resolve it right they can just make up another thing they'll just look at the patterns and be like this is sort of like that that's the explanation and that's that's powerful right you have direct power to minimize free energy right there and then right it's super powerful and attracted so I can imagine What's Happening Here in the way you've just described it is you have this sensory signal which signals that oh your belief was wrong but because my prior are so sticky because they're granted so much Precision I can kind of just ignore that evidence against them and it's not like I'm actively doing that that's just part and parts of the way that I'm constituted good why I think well that that's um it's kind of a bleak picture but actually I think um it it impels us to have some epistemic humility which is that our explain all of our explanations for the world are probably in the service of minimizing complexity um while trying to maintain the modom of accuracy and we're missing out on some some important details and that's true for all of us not only the the so-called Crazy Ones this leads us quite nice yeah go go go go pleas I think it's not just a bleak picture I think it's also hopeful because it gives you a pathway like how do we allow or help people to potentially get out of that yeah um and the way to do that is through kindness through empathy and through a soft approach out you can't you can't bash someone on the head to change their beliefs but you can slowly without making fun of them without without telling them that everything they believe is wrong just slowly untangle one thing you know like be be part of who they are and if you approach this in clinical practice what this means is instead of telling someone your belief is psychotic this is absolutely not real you should more like be like okay well tell me what you see bring me into your belief and together from your perspective let's move out slowly it's it's it's actually something that they do now uh in certain approaches um to to clinical practice with people who go through psychosis and it it

does work it does make people feel better it pulls them away from the edge and it's an epistemic stance that is generally um less pathologizing right it come yeah it comes back to what we're saying it's kind more Humane approach to understanding one another and yeah and and just flows nice ly from like it's we're not making this stuff up this is how the math and the physics unfolds in in active inference um you would slowly change the Precision that you Deploy on certain parts of your model through conversation but as you said it's very rare that you can do that in a single blow so probably not advisable um which actually leads us you know this conversation regarding social norms um the way we group together the way our cognition is distributed right we're not just siloed inside our skulls we rely on one another we offload to one another that brings us very nicely on to a second part of your work which um I feel like because of your background in sexology you've been working on social scripts for a long time um and you have a very deep rooted interest in it so it's going to be really cool to unpick this with you um and yes it is well I I'd like to discuss social scripts and the kind of active inference account of them um obviously when people hear the words script they think of drama the theater films whatever they might not think that their life is scripted unless they they've got some sort of Truman Show syndrome um what again very simple question to start what is a script and what does it say about our Behavior right so there are many again many definition of scripts this is part of the reason we did these studies is oftentimes in social sciences the work is beautiful it's deep it's um again empathetic um but it tends to not perfectly Define its terms or at least not agree on the definition of the terms so many paradigms exist that use the same term and can't agree on exactly what it means if you try to bring it together they will adamantly disagree that these terms can be combined because of fundamental um issues in the uh in the definitions that's something I kind of want wanted to curtail through a formalism and so what we showed is that there were basically two main ways to understand scripts the first way was through weak scripts we weak scripts are the cluster of Concepts that generally exist together and you can sort of draw a mark of blanket around it so um that's the reason that prototype Theory essentially shows well can you really Define a chair like there's many things that can be a chair most things that are a chair do not fulfill what we generally consider to be the Prototype of a chair and lots of chairs don't have backs or lots of chairs uh possibly can be sat on is it still a chair or some would Define it as a chair nonetheless like like there's there's many ways that you can define a chair but in general you know there's like a cluster of elements some of which which are higher probability and other lower probabilities so think of a restaurant it's the it's the example everybody gives in a restaurant you generally

know uh there's probably food you're probably going to sit not every time but probably you're probably going to sit probably food it's probably paid it's not free you know you can just go and get it um and in general but not always there's people giving you the food and making the food for you so that's the you know these are Concepts that are Loosely and more likely or not associated with the concept of restaurant but now there's the strong scripts so inside that cluster of scripts there's a sequence you don't eat right away you're going to enter the restaurant probably going to get seated by someone you don't always pick your seat sometimes you do sometimes you don't but you know you're going to sit somewhere and then you're going to order something you look at something you're order something and then you're going to eat maybe maybe you pay first so you see how the scripts themselves have some Wiggly Ness to them like some parts of the scripts can come before or after but there is generally a sense that something comes before another thing and there's this some of these are higher probabilities so like I have to enter the restaurant before I eat there and I probably have to eat before I pay these are like generally uh generally pretty probability uh in between can wiggle a lot like um maybe I'm going to take 30 minutes to order maybe I'm not even going to look at a menu maybe I'm going to be brought food in the dark like see what I mean like there's there's wiggleless um and that allows for some scripts to sort of um change a little bit this is also why scripts in general tend to be a little bit porous they don't stay the same thing forever so you have scripts that are of a very long temporal scale in the way they unfold and there are scripts that have a very long temporal scale in the way they repeat so for instance a wedding a marriage that's something that's been this existed for a while like we've had marriages for a long time the exact way that marriages unfolded that you know that's changed over time and it's still changing uh and its purpose also has changed to some extent the the people it applied to changed also but in general you know we consider marriages to be between two people who are going to pull their resources together and we assume want to stick around with one another for a while um and so in general we need for a wedding to happen for two people to know each other at least a little bit you know at least once just a little bit they need to go through the ritual of marriage because otherwise that doesn't apply and then they probably need to share some of their resources that repeats over time that's repeated for maybe like I I don't exactly know when the first weddings happen but I know there are marriages in the Bible so you know there's some degree to that um but each marriage lasts like 20 years maybe maybe 40 maybe 60 if you're lucky like you know what I mean like it used to be very short bursts of time now it can be shorter or longer but the repetition of it happens so that that time scale is very long

even though the actual script itself is very short and then there's very long scripts like um like a war will happen over a long period of time or like um the rise of uh The Rise of an Empire generally follows the same kind of elements of a scripts or the rule of a ruler or Etc so these can be very very long um so that's that's a script that's a script good um I think um yes the first thing that I that I can think of when it when when we hear the word script again it's obviously acting to some degree before I get into the question I'm kind of authenticity um there's I think when people hear about mechanistic explanations of behavior they slightly worry that they're being reduced to some automated robot um especially when we talk about something like habits in active inference so um lots of papers talk lots of papers sort of borrow from reinforcement learning um approaches to habits this notion of belief free um habits versus belief-based behavior um or value free value based however you want to formalize it but it kind of comes down to the idea that through repeated practice we arise in a certain context and act in a certain way and under very sort of stringent conditions like if I touch a hot pan um you know my body is reacting in a habitual way I know it might seem extremely reductive to say so but I'm jumping out of the way but I'm automated in that way I I do not choose to um and it seems like in principle one could codify a loss of behavior in that way whether it's opening a door whether it's what to do in a restaurant so how much um flexibility or degrees of freedom it does the active inference framework or or or scriptology I don't know if that's the word theology for scripts um Grant the agent in these social constructs you said that they they they're quite porous they change over time but within the actual synchronic moment where the agent is acting where does their own um agency and and and power to actually act upon the world come from well it depends on where you're positioned relative to the script so everything is perspectival again and let's imagine that you co grain the world into things um you're not seeing every pixel right you're seeing clusters and again these clusters are defined by these weak scripts so how can you recognize something as a door well something is a door because it has a handle and you can sort of see the shape it's not quite the wall but have you ever been say uh to Japan where it's wood panels um would you be able to tell where the door is if there isn't a handle like you'd have to see someone opening the door and then you be like all right that that's that's a door it opens and I can go through like that's that's a door um similarly have you ever been to one of those very fancy fancy mansions um where everything is glass and there's no specific handle so you're like oh God I I want to go outside and it's like it's right in front and you don't know what to do yeah yeah you wait for someone to do it so in that sense that's why scripts are interesting you are going to learn by looking at other people

do things and instead of you figuring out a new way to interact with that door you're just going to repeat that because it's computationally efficient to just you know you have a purpose you want to be on the other side let's just do the same thing but now let's imagine you know there's a door in front of you it's obviously a door and there's someone coming and you're at this awkward distance where it's like if you don't open the door for them it's kind of rude yeah they're not quite there yet so what are you going to do you don't have the right answer there's no proper way to exist here because it's a little bit ambiguous what you're supposed to do you're in the middle of two scripts here you could open the door for yourself and go and risk being a little rude so there's there's a there's a cost here or wait for them and there's a cost of maybe it's awkward because now they feel obliged to run so you're going to weigh the costs of this and you're going to change by your by your decision by your script by what you're choosing to enact here you're going to tip the scale a little bit so in general the way the scripts change is in these uncertain places in these moments of like this hasn't quite been defined there really isn't much of an answer here that is like the obvious one so I'm going to tip the scales and next time depending on the outcome the cost I paid this will act as evidence for the future of you and that other person you have crafted the world through your choice here so I wrote a paper on this with uh Pierre on um on gender essentially explaining in gender seem to be pretty binary like this is not a blanket statement obviously it's not binary every everywhere not across cultures not all the time but let's say in North America let's say in the 50s it was pretty binary right it was pretty clear you if you were a woman you were this if you were a man you were this it seemed like it fit within specific roles or let's say let's say the let's the 30s because the 50s is exactly the time where it started shifting but let's say in the 30s and then the war happened and what happened well you didn't have enough men to do what was need to do so the script that was generally functionally Associated to a role was no longer being able to be fulfilled but we still needed the function and so then we were like well okay what's the closest thing to a man well you know another human a woman woman could do this it's just not in their current role but they can do this and so it's like all right well the script that generally was Associated to women now expanded a little bit because it expanded the porosity also expanded because the the semantics semantics are either um very strictly constricted or they start smashing into each other because if I connect one point everything else can now be connected right so if I can work I can wear pants if I can wear pants I can wear skirts if I can wear skirts I can control my body if I canol my body I can have sex you see what I mean like it it kind of all flows from these natural normally rigid connections but you made one

connection and therefore well now you can make more connections and there was no real functional error when women did that right when women wearing pants it might shock you socially but she's not going to die you're not going to die so it's acceptable you just have to find the right people around whom to do it um and so with urbanization with the fact that after the war uh people gathered in City Senter with industrialization Etc like with you you you had to repopulate you had to find a job you had to etc etc etc etc you ended up with people being able to connect with one another technology also increased and grew uh we had more access to Media more access to each other and through this mish mash you were capable of fostering certain kinds of paracity such that well now gender is you know does it mean the same as it once did do we even really need those categories anymore some people don't even think so anymore it whether they're right or not is irrelevant to them it's now meaningless they they don't need that category to exist in the world some people still do and they use it but we found that the fluidity of gender has been made so overt so clear to people that they feel they can act upon that they feel they have the choice to change the script themselves so I think it starts from there's either a functional need for a switch to happen or there's so much uncertainty that effectively it can go either way so it'll really be about you know your decision It's Kind it's it's a little bit random maybe your specific um position in the world will make it so that you are going to make that choice but somebody else coming from a different perception would have made a different choice so the script in itself was slightly um uncertain at that level cool yeah and I guess the conception that comes to my mind is that one can't think about script evolution in terms of these enormous steps I mean clearly you do have these occasional historical moments that call for radical shifts in Social convention but if you think about the evolution of a restaurant for example there's been no single event at least that comes to my mind which means that people have to wait to be um sat in a European restaurant and maybe they go and sit at their table in American whatever it is but maybe through these kind of incremental changes require you know Necessities pop up in certain niches and are less relevant in others so you you see this kind of staggered Evolution um which brings me kind of question I probably should have started with which is in terms of active inference you spoke a lot about function of um like you know when the function changes so the script will change well that brings me to just ask a very simple question which is what is the function of a script for um free energy minimizing creatures like ourselves what what does it allow us to do and maybe perhaps on the flip side of that how does it constrain us so scripts allow us to avoid Computing everything all the time because you have a script you can make plans much further into the

future you can know how to act um in a situation even though it might be a little novel so you stay relatively stable no matter where you go right there's never I mean it's not true if you have a lot of scripts and you know about a lot of scripts no matter where you go you're probably going to be fine you're probably going to know how to get what you need who to talk to when you need something etc etc so because you have these sort of pre-baked policies that have high certainty that you can recognize because you can understand what triggers that script you're very adapted to the environment and that also allows us to avoid friction so when I know what I'm supposed to do at a restaurant I'm not going to cause a big commotion by doing something ridiculous or by doing something highly unexpected for everybody else we're naturally going to self-organize into the purposes we meant to fulfill when we went there that's why it's an interesting experience to go to a different country and make errors and realize oh wow okay my assumptions were wrong this is not how things are done everywhere there are other ways to do things you loosen the prior over this is the script and you add an extra layer of context yeah this is the script here and this is the script here and your model becomes a little bit more complex but also more adaptable so scripts are superb useful for self-organization and for planning over time Splendid yeah yeah that that's just a super useful addendum um in so far as yeah we are limited creatures if we could compute if we could do perfect Bayesian inference and compute every single contextual element of our world then then then we probably wouldn't need them right we would just act um omnipotently and omnisciently in all context unfortunately well maybe not unfortunately it's probably quite a good thing um you know I was talking about kind of one inherent worry that perhaps all of us have is that we're just becoming the vehicles of our culture that we're acting without really thinking that we're just um yeah that we just embody our Norms Without Really ever having a moment to think well did I choose those did they choose me um this may be irrelevant but I'm curious to see what you think about it um there's an Italian playwright called Pirella Luigi P who was writing in the early 20th century modernist uh play and he wrote a play called um Henry IV and K in Italian and in that um a kind of man acts um they're doing a sort of play in which he is playing the role of a nobleman on a night and he falls off and he ends up believing that he is Henry the 12th century Henry IVth and acts accordingly and shapes the whole world so that because he's very rich originally so that all of his Butlers end up wearing Medevil we and whatever anyway it it unravels during the play that actually for the past five years he's known like they will think that he's mad he's actually returned back to his normal condition but continues to act as Henry IV because philosophically those people are more mad than him because he's

actually chosen his Madness he's chosen his script whereas other people just embody or vehicles for the ideas that are relevant to their context which brings me to something that Yung said which is that um people don't have ideas ideas have people so just sort of throwing it out there all of this kind of a smorgasbord of authenticity to what degree um do you think that active inference agents really are anything beyond the contingencies of their time like do we have anything new that we can bring or is it only um a matter of circumstance when we do something kind of random and and change is there any notion here of like genuine agency or genuine agency I know I know I know I know I know it's full it's loaded that question I think agency is a Continuum yeah it's first of all agency is really hard to Define in Psychology you sort of Define it but if you try to Define it anywhere outside of its functional like observable metrics like they did this and they assigned it to themselves kind of thing um it's really hard to Define but if we were to Define it as say a degree of entropy let's imagine that I expect you to do this when that right so there's like this absolute Baseline that you can be perfectly predicted and then based on that anything you do outside of that prediction is a form of agency like if I don't factor your actions into this you're going to be this but given your actions there's a degree of entropy and uh the more unlikely your actions are the more agentic you might be because you factored into the shift in the decision um let's assume that's a definition but it would be a very contentious one not everybody would agree with that I think we are very agentic we're constantly exploring we're not just exploiting we're not just uh pragmatic we in fact these two terms exist epistemic and pragmatic within the active inference framework where you try to balance between the two you are trying to balance between uh a pragmatic outcome that keeps you alive uh which would be what these people are doing and following the the person they're trying to stay alive and doing what he's saying because they kind of don't have a choice if they want to keep going but to do so they might also be very creative they might improve their model of the world they might be gathering different things they might not act exactly the way he had thought because if he could compute it all then why would why would other people do it you need somebody to gather the knowledge that you currently don't have so there's always a degree to which agents within certain sets are agentic and so that's where we Define those scripted super sets there's highly certain things like who has the money well it's not me it's you so that thing I know if I want the money I know that you're going to tell me to do something but in the way in the something in the distance between the moment that something is fulfilled and me fulfilling it there's like there's there's a million miles it's practically infinite like depending on the granularity of the order as well

there's there's some um space for agency and so what happens is um we sort of agree on what fulfillment means here maybe you know but you didn't necessarily have a very clear picture of what that fulfillment meant you you have some criteria and everything else kind of falls outside and if we were again fully try to Define what those criteria mean we would go in sort of infinite regress of it's impossible to Define everything you need a definition to Define yeah so um because of that like infinite momy because of that space in between the the the time you are given the action and Fulfillment there is space for agency interesting as you said it probably would raise you know raise some contentions from the philosophers um I guess my my push back on that just to sort of play Devil's Advocate would be um this notion I mean just back down to the definition of agency and we don't have to go deep into this um because it's somewhat peripheral but I guess it's interesting to think of it as entropy um away from my expectations of you given my model of you is obviously contingent on my knowledge of you right so if I'm a baby well everyone is super free because they're doing wacky stuff the whole time so I guess the on the complete other end of the spectrum you have a God or thees demon who can model everything you're going to do and in that you're just kind of some pawn in some chess game so completely depend as you as you said it's dependent yeah exactly it's all perspective um which which I guess is actually one of the fundamental insights of active influence which is that um what is free energy to you might not be free energy to me because we're all self-evidencing or self-organizing for for our own ends for for the maintenance of our own models all right excellent um cool well then we we'll jump to the third paper the third theme but it actually all kind of ties together I think which which makes sense given that obviously it's it's your work and these are coherent uh you know coherent interests um you wrote a paper I think it was this year with um with Ali Ali ramju who's obviously is a great friend of The Institute um and and helps Daniel out with with everything so shout out to ali um and this one really piqued my interest um so this is intraaction inference um I think right now it's am I right to say it's only out on Sai archiv at the moment is that right m uh yes it's the preprint it's still the preprint well definitely people should check out because it's a it's it's a great paper um they might have to get their head around some terminology uh because it's quite rich with with with specific terminology but I'd love to be able to unpick some of the insights that you and ali uh came uh came up with in that paper I guess the first one here is that you say that um and I can quote here a genuine Fe so Fe is free energy principle a genuine Fe based ontology must reject any discontinuity between matter and subjectivity as FP does not confer any uniqueness to cognitive or active agents the same mathematical technology

applies equally to a whole range of entities from inert rocks to the brain so there's a natural tie there to what we've just been talking about in terms of agency and free will but I want to actually stay a little bit away from that and talk more about this discontinuity this um this separation that we in sort of intuitively make between matter and mind or matter and subjectivity just for the kind of um Layman people out there go well of course they're different because this is all mental and the chair in front of me is not and okay um so what does this so This sounds bizarre for them what does it mean to reject a discontinuity between matter and subjectivity right so let's start from the beginning please materialization is the key here matter isn't static it's constantly becoming it's shaping itself it's being shaped and basically it's that's that's how what we were discussing which is neomaterialism is distinct from materialism where materialism presupposes elements outside of say the perception of the individual it presupposes it there is a true outside in the world and you have to do your best to figure out what is that truth in NE materialism we're saying it's not that simple um we have basically um a relationship between what is being perceived the perspective The Observer and what is being perceived at all like like this this um the element outside is really just um a a relationship between two things becoming together um so basically we we try to recognize that there's an indeterminacy of matter and um the this this move movement of matter is constrained by possibility spaces um so like we discussed like the scripts right scripts are possibility spaces but they are also constrained by say um physical constraints like you can't fly like that's just not a thing yeah um but you know um there are perceptions that shape what you consider to be things around you and you will shape the things based on what you consider to be these things um so basically given these possibility spaces you're going to create metastable formations and you everything is going to align around these metastable formations it's the niche formation idea right and so materialization in this sense is um the iterative process where matter continually redefines limits and boundaries these are constantly changing and being um negotiated and this negotiation is due to the way that things um self-evidence right so matter being relational and in a sense self-caused because things self-organized is moving this um this division between the ontology um that aligns the view of um the self and the matter here Everything is self-evidencing Everything is self-organizing and effectively um you you you are also constituting others and they are constituting you others or other things so the the the boundary here is virtual it's dependent on a a third sort of perspective that defines that you are not the thing in front of you um the um this is where we basically determine that there's a probability of sensory State that's interpreted as evident for the agent's existence uh and so

everything around you both uh is evidence for your existence and you evidence as well right does that so far is that is that clear yeah yeah um just for those who maybe are more familiar with the terminology and active inference am I right to say that this possibility space could be uh defined as an attractor set um that is that bit of a blanket statement uh no I think that's that's interesting so basically um in the possibility spaces or at least according to Thea and the uh you have virtual and actual uh properties and capacities uh and so if we relate them to the Dynamics of fe uh and active inference um you basically just have a temporal depth of these possibilities that are um directly afforded to you or potential given say a new agencing of the the elements in play and so yes you have farther away attractor set that are more or less likely given the complexity that would need to be given that would need to give rise to them so if you think of energy energy is just um the the the improbability of something happening high energy means it's highly unlikely for it to happen um and so um when we when we deal with these uh possibility spaces we're basically talking about um co-constitution of what might happen and what is happening given the collapse of possibilities by what you are doing and what the world is doing to you so you're constantly in this sort of [Music] um twister in this twister where you act on the world and it acts on you and it's constantly being uh being shaped right and I guess an important elision there not only between subjectivity and matter but also ontology and epistemology so ontology this this the study of what is and the epistemology the study of how we know what that is or what it is in the sense that a free energy minimizing agent is in the game of basically getting to know themselves in some strange way so they're garnering evidence for this own implicit model that they have of themselves and the world that they want um given this attractor state so that's a really beautiful way that to know oneself is to be oneself or vice first and I Know Carl has some strange inversion of the cartesian um I think therefore I am this is this is a really interesting question because I think uh or it's a really interesting area because it yeah I mean just brings up so many fundamental questions one being that I think in your paper you speak about how one can never perfectly actualize a or or be static in the possibility um set or or put in terms of attracting States you could never be at stasis where you could never be fully attracted to your attractor set um you're always uh in well at least for us we're in this kind of itinerant state where we uh are open to the world and therefore has to deal we have to deal with its kind of stochastic nature um I guess maybe I'm being an idiot what would the World maybe um what would the world look like if we managed to reach that stasis so this I I understand that this Dynamic um unfolding this um this mutuality of self and other this getting to know oneself what we're really coming to is the

notion that matter as a process rather than as this kind of rigid static thing but if that I mean if that process essentially came to a halt are we invoking that matter itself recedes the matter hides matter if matter is processed once this thing stops if it reaches its attracting state if I reach perfect homeo States whether do I stop is my ontology rooted in my dynamism or is there something over what was the any materialist say about that or is there something over and Beyond this process of self- evidencing no I think you touching upon it I think if there's no more energy if there is no more information being exchanged there's nothing happening and therefore if nothing's happening me you're dead is there a strange Paradox there that from the outside right like if the claim is that this extends existential imperative is to minimize free energy to try and get to this precise level of homeostasis it sounds like what we're trying to get to is equilibrium right like it does I just want to stop right like why am I cursed with this fact that I can't right that like I'm cold and H and off you know but like that seems to be the goal but paradoxically it would all stop if that was fulfilled right yeah that that's kind of the way that I'm interpreting what you're saying so again i' careful here I'm not giving a Tey to something that doesn't have a telos um but how how do you kind of unpick that Paradox yeah well it's because this isn't we're not meant to fulfill it right we are the way we are because there because the universe is the way it is like the universe leads towards entropy right eventually there is no more energy eventually everything is spent eventually everything is equivalent there's there's no information to be gathered anymore that's just how things are and it and it exists on such a temporal scale on such a massive temporal scale that at the level of us this is and always will be so we are not meant to ever reach a state where there is no information to be exchanged where things don't change anymore so obviously we are constantly pulling in the other direction and it kind of looks like we're trying to reach a point where we no longer move but that will never happen right so it's natural that if the slope is like this you're trying to go like this and and basically what you're saying yeah but what if you reach the top of the mountain what happens then well yeah if we reach the top of the mountain eventually we would cause our own issue right we would we would um we would reach the point we're trying to avoid reaching we'll never really get there and at the same time consider um we are a product of this system because the system is ongoing so we also contribute to reaching this point we are consuming energy and we are going to help eventually reach the heat dep of the universe this is not like obviously we're we're trivial at this point right we're a blip but imagine we got better at it then we would probably just make it much much much faster than it would have been otherwise that's what a lot of people think that's what well in

many ways that looks like what it kind of looks like what we're doing yes I um yeah maybe I need to just banish this romantic notion that there might be some top of the mountain um my sopian struggle will actually have a have some benefit you know some positive outcome I think something that you mentioned there in passing which is really important and really interesting and might not always be made explicit to those who are learning about active inference is that we're talking about information here so I'd love for you to um be able to I I'd love to hear a bit more about information um we think of Mata again as this kind of static entity the models that we we learn at school um but at least to my very Layman ears we have this with um free energy principle and just modern physics as it's unfolding this really beautiful convergence of matter and information and you have uh statistical mechanics and you have stuff like Shannon information and all of these really fascinating Concepts when you say that we will get to a point where there is no more information trans er um where basically yeah just everything is as is um again I just digging into the ontology there is that to say that and I know there's no hard and fast answer is that like are you postulating some kind of it from bits as David Chalmers would say that like kind of the fundamental unit that we're talking here is information and matter is in some way emerging from that is is there some ontological hierarchy ontological structure that you can point to there or is it fuzzy so that's an active area of research obviously there is no perfect answer in the relationship between information and matter like we can always point to the land hour principle you know uh we can talk about um how um mass and energy are related and if you have energy then you have a sense of information because effectively um information is related to the concept survival where like the negative log probability of an observation becoming true uh is effectively you know surprisal and therefore information so there's a relationship there right and if what we're talking about is the relationship between um you subjectivity and therefore perception and uh or at least Observer Dynamics and matter which are mutually shaping their is a sense in which the way that information is is is exchanged effectively shapes the boundary which gives rise to what we can consider um the ontological scale at which matter exists so in a sense yes but really when we talk about information we're really talking about a degree to which um a certain bound on a surprise is related to a sensory data in a model of the world so um information is negentropy when you have unlikely something that's information if everything is equally likely then there's no information there right so that's where we get to this relationship between the heat death of the universe and the absolutely equivalent um probabilities for anything because there is nothing there and the fact that uh if you reach the

point where you weren't exchanging any information effectively uh you you there's nothing going on there there is no information to exchange yeah I think that that sheds light on um a potential point of confusion that people might have when they read the active influence literature which is that when we're talking about self-information we are not talking about a kind of self-evidencing result maximal self-information is not self-evidencing in fact it's quite the opposite um so that's that's that's useful to remember that the in like if everything was unfolding predictably you would have less information right like your model would be working optimally if you're getting more information out of your model well then there's some Divergence going on here and so so so yeah it can all be kind of cast as this um quite simple statistical picture excellent good well M the last thing I want to speak to about I mean this was fascinating and honestly on this topic I would love to just go on forever um because you've also got another um excellent paper that people should check out which is about Consciousness um and holographic screens and so there's plenty there's plenty for people to get their teeth into and we've really barely touched the surface but I guess the last thing that I want to speak about in some depth is your work at verses um um and artificial intelligence and and your PhD as well of course which is in cognitive Computing so obviously within the um the limits of disclosure and how much you're allowed to talk about maybe you could kind of give us a little bit of information pardon the pun about what versus is what versus does um and how it's kind of revolutionizing intelligence in some ways um and and how that's being embodied or encoded into artificial agents right so I guess I can give you the key difference most people consider just like we discussed that intelligence is a property of an inside and that everything else is outside what we consider is that intelligence is a phenomenon it happens as a function of the interactions between different kind of um individuals agents objects there is information being passed and it's the integration of this information that can be self-reflexive through different kinds of boundaries that enable acting in a specific kind of way relative to an outcome you wish to see happen so you acrw agency uh given how much information you can sort of pull together so our approach to intelligence is shared intelligence and um we consider that shared intelligence is not just about shared bits of data it's also creating perspectives forcing and embedding into the world and because you have an embedding into the world you have the possibility to dissolve and when you have the possibility to dissolve and information you have veilance when you have veilance you have emotion and then you have the possibility for empathy you have the possibility for theory of Mind coordination cooperation and eventually shared intelligence these are key elements that have not been used in the literature

so far at least are not being um delved into as much from say the machine learning uh crowd they're doing amazing work they've allowed for very deep U processing of data but this data is always decontextualized the key issue currently is llms hallucinate and they hallucinate because they have no clue what the context really is or what you're trying to do really and where they are relative to this context they have no Loop and more than no Loop they have um no true sense of how to coordinate with you they don't know if something happened they they don't know if you're happy with the answer really like you can give a thumb up or a thumb down to Chad GPT but while it's giving the answer it doesn't have that so all of these elements together make it so that U potentially this approach is going to hit a wall our approach is more natural and gives rise to this fundamental principle of mutual information exchange where in order to improve your model of the world you have to get to know the thing in front of you better you have to integrate part of it as a part of you and now you're shaping each other so this whole alignment problem sort of falls out because it's no longer one about One True Value alignment it's about how all of these pieces coordinate together and create create a harmonious self-organizing system that reaches towards something that resembles a metastable state awesome so just to kind of backtrack there and and maybe give a a broader picture um really good that we now have a kind of understanding of of of what versus is trying to implement I thought the mention of kind of the machine learning LW and the llms is really useful here because people have I think have in a notion now because there's been so much news about chat gpt and Bard and neural networks that that's what artificial intelligence is perhaps you could give us a very simple um carving up of the space in which if you could a very brief sort of synopsis about how what an llm is what a large language model is how it works what you know like vector spaces and probabilities and stuff like this and then what an active inference system might look like um whether that is embodied or not so I might be leading you down a down the path yeah so essentially uh Transformers or attention mechanisms that have self attention and attention to uh the text in question they embed different uh relationships between different different words that give you essentially a probability of a next word being produced uh it's really that simple it's a it's a hierarchy that retains certain words and then gives you probabilities at different scales so the more complex your Transformer hierarchy the more complex your attention can be but effectively it's just that it's what is the probability of this word given that that word and that word and that word were said so it's content to content really it's not it doesn't understand the structure of the world it doesn't have a model what it has is just a sort of given this word that word given that word that word given that word that

word and it's very performant because knowledge the crude knowledge that we have grafted in language has a sort of underlying model you know like we don't talk nonsense most of the time when we talk we try to make sense we're not just spouting out a bunch of random words out of nowhere that speak to nothing of our reality but that's limited to that our approach basically gives our models the capacity to from what it observes create another layer and that layer tries to understand what the causes are why this happens not what word preempted that word but why was that word used and in that case what you would get if you created an LLM with an active inference agent what you would do is it's not just something that predicts the next word it's something that predicts the cause of the next word and therefore has a sense of why you're saying what you're saying what you're trying to do and what it might say if it chooses to go along with that perspective um another difference is that the ELMs are not really placing themselves in that Loop it has self attention relative to words but it doesn't know what it is right it doesn't know what its actions are it can't make a choice relative to an objective that's something that you get if you have an agent that places itself within the loop and understands the outcomes of its own actions relative to these uh to these causes great yeah yeah that's a really useful um separation there distinction between the two um an interesting aside that we could get into but maybe it's maybe down the time it's not worth it which is about the degree to which language does indeed refer to some external world because you have all the external you know the semantic externalist semantic internalist I'm sure Mr Chomsky would have something interesting to say but actually I'll refrain um something I have heard which I think is bit more interesting and I was really happy that you mentioned the action perception Loop that you are encoding but maybe a better word there would be that you're embodying or you're allowing these agents to embody and I really want to have just a moment to speak about embodiment because these large language models don't have sensory States they don't have active States and for anyone who's seen a Markov blanket that seems pretty fundamental to Adaptive intelligence to being able to enact self-organization in the way that we do it um something that I have heard is that in some ways this embodiment grants us and perhaps the robots in the end with a sort of care um and I mean care they're in the hiigaran sense there there's a kind of um there's a tendency to lean towards the world in the way in which we're interested in the way it's coupled to us and the way it interacts with us and the way we interact with it whereas if we are not embodied exactly because you don't have that causal impact on the world um you don't you don't embody that fundamental phenomenology of care um it doesn't really mean much to you to put it in s of layman's terms I

guess bringing it back to some a topic that always I want to speak about and need to stop speaking about so much is consciousness it's a very loaded question but is there a degree to which embodying an artificial intelligence doesn't just endow it with um goals that are just in principle more align with ours but an actual sense of phenomenology a Consciousness and if not why not and if not what else would we need to add and again I'm not expecting you to solve the hard problem in 10 minutes but just what is the kind of active inference perspective on a question like that right so I think we need to reformulate what embodiment means I think embodiment is just a relation to self within a context right and so because because why would a body just mean this yeah are you less embodied or you're in when you're in the virtual world I would argue you actually are because in effect your perception of your own body is just an illusion created by your brain so what is embodiment really I think embodiment is that it's a relationship to self relative to context which means there's a boundary somewhere if you are the entire context there's no boundary so there's no real embodiment you have to somehow separate separates something now maybe you separate your own Parts maybe you are every part but you understand how the different parts are different from one another that's maybe one thing right if you if you imagine the the scenario of a say an Overlord AI that is connected to everything and everyone its embodiment might be how it can pull different strands of perspectives given its different pieces for instance so we have to rethink our understanding of embodiment to be a little less centered on our own experience and this leads us to isometry why would phenomenology only be given to something that resembles a physical because I imagine that's what you kind of had in mind right you had sort of in mind like a humanoid sort of Android thing which we we would give to an AI and therefore they would be anchored in the world in the same way that we are and if effectively that would give them a phenomenology but I think if we're thinking of embodiment we're really just thinking of how the thing the agent relates to itself relative to its boundaries and how those boundaries Define how information is processed relative to its objectives I think the problem with that is that it doesn't ensure that we will understand each other what we want is to endow the AI with the possibility to truly grock us and for us to truly grock it we need to be able to exchange this information in a way that's you know useful um so we tend to want to give it an embodiment that resembles ours I do think embodiment in the sense that I defined it is required for what we understand to be Consciousness because you need the boundaries to reflect each other without the boundaries reflecting each other you have a form of information being integrated but you don't have self and therefore you don't have self- reflection and I don't think given that you

necessarily have um access and agency over that access you just have you know movement along some gradients um so that's what I think will be required to give a form of Consciousness to um AIS that have a version of embodiment but just to clarify are you saying that a um recursive or embedded system of markov blankets in which information is being exchanged up and down and in which errors are being passed up and um temporally slower levels contextualizing these higher order levels are you saying that is sufficient for Consciousness I understand that it might be necessary for Consciousness but something that I've heard you and Maxwell speak about actually on with the active inference Institute is this notion that the that Consciousness might entail some layer of that cake which is so-called right only so it isn't being contextualized it it it's just I don't know what issuing information how would you what does that model look like um and have I well yeah um I I I don't want to completely butcher it so I'll pass it over to you I think the right only parts of the models are just they're dependent like um depending on the scale everything can be right only like it's it's really a question of what is the degree of access and I think when we Define right only we Define the conditions for the thing to survive you don't have a choice but to breathe air right you don't have a choice but to drink water these are not um choices you're making so to some extent there is one layer at which something is like it has to happen right um and therefore it's not something that um that is written on because it's likely to change so at the level of an AI That's what I mean by the perspectival aspect of their embodiment with a boundary if they have a boundary some boundary whatever that boundary is they have a condition for that boundary to be dissolved and that condition those conditions Define the right only uh layers of the boundary does that mean in some sense that being able to for the system to recognize it's distinct from something else from everything else is a prerequisite condition for Consciousness within this model that's right that's right in order for an a thing to have a self and an embodiment it has to be somehow bound from other things like it it has to be different to some extent or it has to be able to take perspectives that enable this difference like I said if you had this Overlord that can pull into different consciousnesses it would still for a time being be able to pull this specific boundary and embody it in a sense yes yes and that and that feeds back very nicely into this Hy doarian notion of care to take a stance on something I think is a way that he might Express that wonderful I mean I I guess my final comment on this would be what given this given that a kind of um the perspectival the the self versus other distinction is so at least in this model crucial to a sense of what it is like to be anything how much Credence or how do you rejoinder those who speak of um boundless Consciousness

experiences or um yeah ego death or Oneness are these um delusional do they or do they invoke some minimal sense of self where do they fit into this story so I think that's a really interesting topic and I'm not someone who's going to um invalidate anyone's experience so obviously again take what I'm saying with a grain of salt your experience trumps my perspective on your experience what I think happens is you're not actually dissolving right you are dissolving certain layers of what generally constitutes yourself and that experience is powerful and transcendental and because you feel like what usually constitutes yourself is dissolved you feel like you're connected to everything else but you're not you're still you you're still embedded in your body right these practices are conceptual and tap into what we believe to be what that would be like but the truth is you don't know no one who's ever reported that experience wasn't in a body so I think we all tap into that same experience when we go through something that resembles ego death in the sense that there are layers that become um Nill like we dissolve these layers but not all layers which means on a fundamental level that infinite Consciousness is still being tapped into from a perspective awesome yeah I mean um I guess I ought to point people to where this philosophical Foundation is and people should definitely check out Thomas metzinger um and something that's really important to point out here is that when we're talking about self one ought to keep in mind that there's a distinction here between the so-call ontological self this kind of entity that I think people instinctively associate with a kind of cartisian Soul this unchanging entity and perhaps what we're more concerned with which is the so-called phenomenal self model how the system um models itself how it reflects on its own being how it takes a stance on its own being and what it conceives of itself and perhaps the attenuation of some of the self- modeling is exactly what's happening in psychedelics and here we speak to people like George Dean who I'm trying to get on and I think that would be fascinating and some of the work that I'm working on is is completely to do with self modeling so so if this is all very neatly tied into current work going on in active inference M this has been awesome um I know I know our time is basically up um where what so what what do people have to look forward to on your side what what are the kind of interesting things that you're working on where can people find you uh where can people reach out to you if they're exhilarated by the stuff we've spoken about today and want to find out more so I'm working on several papers one uh on uh emergence in uh in active inference I'm also working on a paper on self-esteem and how to model it in social groups and I'm working on a paper on sustainability and how uh that rises from the notion of resilience and mutually uh effective resilience um you can find me on LinkedIn on Google

Scholar uh and uh sometimes on Twitter very very rarely I would much more encourage people to reach out to me U by email if they find it on one of my papers or on LinkedIn directly awesome thank you Ma this was great thank you